AI-Based Electricity Demand Intelligence System for Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Electricity demand forecasting is a crucial element in ensuring the efficient management of energy systems. In Sri Lanka, accurate short-term load forecasting (STLF) is vital for balancing the electricity grid, especially in the context of peak demand periods. Traditional methods, including statistical and machine learning models, are effective for predicting demand but often lack the ability to explain the factors influencing demand fluctuations. This lack of transparency creates a challenge for energy planners and operators when making real-time decisions. The objective of this research is to develop a behavior intelligence module that analyzes electricity demand patterns by identifying similar days, detecting anomalous demand, and recognizing peak-demand events. The proposed system utilizes unsupervised learning techniques like K-means and DBSCAN for clustering demand profiles and anomaly detection algorithms to flag irregular consumption patterns. This component will then label demand days with meaningful categories, such as "normal weekday," "peak-demand day," or "abnormal anomaly day." By offering insights into the underlying behavior of electricity consumption, the system provides transparency and helps planners make informed decisions regarding load management, optimizing resource allocation, and improving energy efficiency. Additionally, it supports improved decision-making during critical times, such as peak-demand or anomaly events, and enhances the overall stability of the grid.

Keywords: Electricity Demand Forecasting, Behavior Intelligence, Anomaly Detection, Peak-Demand Events, Unsupervised Learning, K-means, DBSCAN.

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TABLE OF CONTENT

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 7

LIST OF FIGURES 8

LIST OF ABBREVIATIONS 9

1.INTRODUCTION 11

1.1 Background and Context 11

1.2. Literature Review 13

1.3. Research Gap 16

2. OBJECTIVE 18

2.1 . Main Objective 18

2.2. Specific Objectives 18

3.METHODOLOGY 20

3.1 Phase 1: Input Data and Preprocessing 20

3.2 Phase 2: Clustering for Demand Pattern Discovery 20

3.3 Phase 3: Anomaly Detection 20

3.4 Phase 4: Behavior Label Generation 21

3.5 Phase 5: Visualization and Output 21

3.6 Phase 6: Validation Strategy 21

3.8. Tools and Technologies 22

4. HIGH-LEVEL SYSTEM ARCHITECTURE 23

4.1 Component Overview 23

4.2 Module Structure 23

4.3 Data Flow 24

4.4 Integration with Other Components 24

4.5 Scalability and Security Considerations 25

5. USER REQUIREMENTS 28

5.1 Stakeholder Identification 28

5.2 Functional Requirements 28
5.3 Non-Functional Requirements 30
6. COMMERCIALIZATION PLAN 32
6.1 Target Market and Customer Persona 32
6.2 Value Proposition 32
6.3 Revenue Model 33
6.4 Cost Estimation 33
6.5 Pricing Strategy 33
6.6 Competitive Advantage 34
6.7 Intellectual Property Considerations 34
7. BUDGET AND JUSTIFICATION 35
8. WORK BREAKDOWN STRUCTURE (WBS) 38
9. GANTT CHART 41
10. REFERENCES LIST 42
11. APPENDICES 44

LIST OF TABLES

Table 1 Tools and Technologies Used in the Behavior Intelligence Module 22
Table 2 Functional Requirements of the Behaviour Intelligence Module 28
Table 3 Non-Functional Requirements of the Behaviour Intelligence Module 30
Table 4 Budget and Justification for the Behaviour Intelligence Module 35
Table 5 Work Breakdown Structure and Timeline for the Behaviour Intelligence Module .. 38

LIST OF FIGURES

Figure 1High-Level System Architecture of the Proposed Electricity Demand Intelligence Platform 26
Figure 2 Behaviour Intelligence Module Individual Component Architecture..... 27
Figure 3 Project Gantt Chart — Semester-wise Timeline with Milestones 41
Figure 4 Clustering of Similar Demand Days: Daily Load Profile Clusters Identified by K-Means and DBSCAN..... 47

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SHAP	Shapley Additive Explanations
LIME	Local Interpretable Model-Agnostic Explanations
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal Autoregressive Integrated Moving Average
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
ML	Machine Learning
CSV	Comma-Separated Values
KPI	Key Performance Indicator
PDF	Portable Document Format
API	Application Programming Interface
GUI	Graphical User Interface
IoT	Internet of Things
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
KMeans	K-Means Clustering Algorithm
RNN	Recurrent Neural Network
XGBoost	Extreme Gradient Boosting
RF	Random Forest
LTLF	Long-Term Load Forecasting
API	Application Programming Interface

F1-Score	F1-Score (for model evaluation)

1.INTRODUCTION

1.1 Background and Context

Electricity demand forecasting plays a vital role in keeping modern power systems stable, reliable, and efficient. For the Ceylon Electricity Board (CEB) and energy stakeholders across Sri Lanka, getting this right is not simply a technical goal. It is an everyday operational requirement. The country faces a unique combination of pressures: a generation mix dominated by hydropower that is directly tied to seasonal rainfall, ongoing dependence on imported fuels for thermal backup, and a consumption landscape shaped by economic growth, climate patterns, and the habits of millions of households and businesses. When these factors collide during a dry season, a fuel shortage, or an unexpected spike in demand, the national grid comes under serious strain [10][13].

The real-world cost of poor forecasting is not abstract. When demand is overestimated, fuel is burned unnecessarily and generation capacity sits idle. When demand is underestimated, the result is load shedding, blackouts, and disruption to everyone who depends on a reliable electricity supply. Sri Lanka has faced exactly these situations in recent years, with recurring power shortages that have affected homes, factories, and businesses across the country. These episodes make it clear that the existing approach to demand forecasting needs to be smarter, more transparent, and more capable of anticipating what is coming [11][12].

Globally, the field of electricity demand forecasting has moved through several generations of methods. Early approaches using Multiple Regression Analysis (MRA) and AutoRegressive Integrated Moving Average (ARIMA) models provided a starting point but were limited in handling the nonlinear and multi-factor nature of electricity demand. Machine learning methods including Artificial Neural Networks (ANN), Support Vector Machines (SVM), and deep learning architectures such as Long Short-Term Memory (LSTM) networks have since pushed predictive accuracy significantly higher. Research consistently shows that electricity demand is shaped by a complex mix of seasonal cycles, temperature, calendar effects, and consumer behavior, making it a genuinely difficult time-series prediction challenge [8][11].

Yet even as forecasting accuracy has improved, a critical gap has remained largely unaddressed. Most systems are built to answer one question: what will demand be tomorrow? They are not built to answer the question that often matters just as much for operations: why is demand behaving this way, and what kind of day is this? Advanced deep learning models can produce highly accurate predictions while remaining completely opaque to the human decision-maker who needs to act on them. For grid operators and energy planners, a number without context or explanation has limited practical value [11].

A further gap exists in how existing research handles the relationship between demand pattern analysis, anomaly detection, and behavioral understanding. Studies on anomaly detection have largely focused on identifying fraud or irregular consumption at the individual

consumer or smart meter level [2][5][9]. Clustering research has mostly been applied as a preprocessing step to improve forecasting accuracy rather than as a tool for understanding demand behavior in its own right [3][6][7]. No unified framework currently exists that brings these capabilities together, grouping similar demand days, detecting unusual events, identifying peak demand conditions, and labeling each day with a meaningful behavioral description, in a system designed for operational use in Sri Lanka.

This research addresses that gap directly. The proposed Behavior Intelligence Module forms a core component of a Short-Term Load Forecasting (STLF) system designed for the Sri Lankan electricity context. The module processes historical 15-minute interval demand data, builds daily consumption profiles, and applies unsupervised clustering and anomaly detection techniques to reveal the recurring patterns and irregular events within that data. It then generates interpretable behavior labels including Normal Weekday Demand, Peak Demand Day, Holiday Demand, and Abnormal Demand Day, giving electricity planners and system operators a clear, day by day understanding of how demand behaves and why. This is the layer of transparent, actionable intelligence that current forecasting systems do not provide, and that Sri Lanka's electricity management genuinely needs.

1.2. Literature Review

Electricity demand forecasting and consumption behavior analysis have become increasingly important research areas as modern power systems grow more complex and the need for intelligent energy management becomes more urgent. Across the world, electricity providers face the challenge of balancing supply and demand in real time, managing generation resources efficiently, and preparing for sudden shifts in consumption caused by weather changes, economic activity, and social events. To address these challenges, researchers have developed a wide range of approaches spanning traditional statistical models, machine learning techniques, clustering algorithms, and anomaly detection frameworks. Each of these approaches contributes to a different aspect of understanding and predicting electricity demand, yet no single existing system brings all of these capabilities together into a unified, interpretable framework for operational decision support.

In the Sri Lankan context, these challenges take on added significance. The national electricity system managed by the Ceylon Electricity Board is heavily dependent on hydropower, making it sensitive to seasonal rainfall variability and the effects of climate change. Dependence on imported fuels for thermal backup generation introduces further vulnerability to price fluctuations and supply disruptions. Combined with a rapidly growing economy and shifting consumer behavior, these factors create a forecasting and demand management environment that demands more than simple numerical predictions. What is needed is a system capable of understanding demand behavior at a deeper level, one that can identify recurring patterns, detect unusual events, and provide planners with clear, interpretable insights to support operational decisions.

1.2.1. Traditional Forecasting Methods

Early electricity demand forecasting relied heavily on statistical approaches such as AutoRegressive Integrated Moving Average (ARIMA), Multiple Regression Analysis (MRA), and Exponential Smoothing models. These methods analyze historical consumption trends to estimate future demand and were widely adopted due to their simplicity and ease of interpretation. Despite their usefulness, review studies on electricity demand forecasting methodologies have consistently noted that statistical models struggle to capture nonlinear relationships and complex consumption patterns, particularly when demand is influenced by multiple environmental and behavioral factors simultaneously [11].

1.2.2. Machine Learning and Deep Learning Approaches

To address the limitations of traditional statistical methods, researchers introduced machine learning and artificial intelligence techniques into electricity demand forecasting. Methods including Artificial Neural Networks (ANN), Support Vector Machines (SVM), and deep learning architectures have demonstrated stronger capability in modeling complex nonlinear relationships within electricity consumption data. A study conducted specifically on peak electricity demand prediction in Sri Lanka compared MRA and ANN models using Ceylon

Electricity Board data. The ANN model achieved a Mean Absolute Percentage Error of 2.50 percent compared to 4.72 percent for the MRA model, confirming that machine learning approaches better capture the nonlinear patterns present in real-world electricity demand data [13]. A meta-analysis examining forecasting methods applied across Australia, Pakistan, and Sri Lanka further confirmed that machine learning and hybrid approaches generally outperform traditional statistical models. However, the same study highlighted that many forecasting models fail to sufficiently incorporate external factors such as weather conditions, climate variations, and behavioral influences, which limits their practical accuracy in real consumption environments [12].

1.2.3. Clustering for Demand Pattern Analysis

Clustering techniques have been widely applied to analyze electricity consumption behavior by grouping data points with similar characteristics. One study applied K-Means clustering to classify electricity load patterns from the Indonesian Jawa-Bali power system into distinct categories based on peak load characteristics. The results successfully identified demand variations caused by national holidays and changes in social activity, demonstrating how clustering can reveal meaningful behavioral patterns within electricity demand data [3]. A separate study investigated unsupervised clustering methods for grouping industrial electricity demand profiles using data from Chilean manufacturing plants. Techniques including PAM, k-shape, and DBA algorithms with Dynamic Time Warping distance measures were applied to generate representative load profiles for demand-side management planning [6]. While both studies confirmed the value of clustering for electricity demand analysis, their scope was limited to industrial profiling and monthly peak-load classification rather than day-level behavioral analysis of electricity demand.

1.2.4. Clustering Combined with Forecasting

Several studies have explored combining clustering and forecasting techniques to improve prediction accuracy. One study grouped households with similar electricity consumption behaviors using K-Means clustering before applying forecasting models at the cluster level, producing lower forecasting errors than direct system-level forecasting [4]. A related study used time-series clustering with K-Means and Dynamic Time Warping on smart meter data from residential households in South Korea, then applied multiple forecasting models including ARIMA, NARX, and DSHW to each cluster. Cluster-based forecasting consistently outperformed non-clustered approaches across all tested models [7]. Despite these contributions, clustering in both studies was treated only as a preprocessing step to support forecasting accuracy. Neither study produced standalone behavioral insights or interpretable day-level demand labels for operational planning purposes.

1.2.5. Electricity Consumption Behavior Analysis

Research has also examined electricity consumption behavior at the user level using machine learning techniques. A study applied Spectral Clustering and Hidden Markov Models to analyze consumption patterns within a smart community, successfully identifying distinct user behavior groups based on electricity usage characteristics. These behavioral insights were then used to design targeted marketing strategies and energy saving programs for electricity consumers [1]. While the study demonstrated the potential of behavior-based consumption analysis, its focus remained on individual consumer segmentation and commercial strategy rather than system-level demand behavior intelligence relevant to grid management and operational electricity planning.

1.2.6. Anomaly Detection in Electricity Consumption

Anomaly detection has received considerable research attention as a method for identifying faults, inefficiencies, and fraudulent consumption patterns in electricity data. One study proposed a machine learning-based Anomaly Detection Framework using symbolic data representation combined with SAX-CART and SAX-MLP classifiers to detect abnormal consumption patterns in building electricity data, providing energy managers with interpretable tools for identifying irregular usage and system malfunctions [9]. Another study developed a hybrid framework combining unsupervised SR-CNN and martingale-based anomaly detection with supervised Two-Class Boosted Decision Tree classification, applied to smart meter data from approximately 4,300 Irish households. The system achieved 90 percent accuracy and an F1-score of 0.894 in identifying suspicious consumption behavior and potential electricity fraud [2]. A more recent study introduced a VAE-DBSCAN hybrid method that combined Variational Autoencoder feature extraction with DBSCAN density-based clustering to detect abnormal electricity consumption points, improving data quality before downstream ARIMA forecasting by removing contaminated readings that would otherwise distort prediction results [5]. Although these anomaly detection frameworks demonstrate strong technical performance, they are primarily designed to detect abnormal data points or fraudulent consumer behavior at the individual meter level. None of them integrate anomaly detection with broader system-level demand behavior analysis or provide day-level behavioral insights for energy planners and grid operators.

1.2.7. Limitations of Existing Systems

While existing research has made meaningful progress across forecasting, clustering, and anomaly detection, several important limitations remain. Most forecasting models are designed purely to produce a future demand value and offer no explanation of the behavioral patterns behind that prediction. Advanced deep learning models such as LSTM and MLP operate as black-box systems, making it difficult for system operators and energy planners to understand why demand behaves in a particular way on a given day [11]. Clustering methods in existing studies are applied mainly as a supporting step for forecasting improvement rather than as an independent tool for analyzing daily demand behavior [4][7]. Anomaly detection

approaches are focused on identifying abnormal consumption at the individual consumer or meter level and do not provide system-level behavioral insights that could help grid operators understand broader demand variations [2][5][9]. Existing electricity management tools and forecasting platforms also lack the capability to classify demand days into meaningful behavioral categories or generate interpretable daily labels that can directly support operational planning and decision-making. These limitations collectively highlight the need for a more comprehensive and integrated approach to electricity demand behavior analysis.

1.3. Research Gap

Existing research in electricity demand forecasting, clustering-based consumption analysis, and anomaly detection has made notable progress in improving prediction accuracy and identifying abnormal consumption patterns. However, a careful review of the current literature reveals that several critical capabilities remain missing and that existing approaches are insufficient to meet the full demands of intelligent electricity system management, particularly within the Sri Lankan electricity context.

Forecasting models including ARIMA, MRA, ANN, and deep learning architectures treat electricity demand purely as a numerical time-series problem, producing predicted demand values without offering any explanation of the behavioral patterns, recurring demand cycles, or unusual events that drive those values [11][12]. Advanced deep learning models such as LSTM and MLP further compound this problem by operating as black-box systems, making their outputs difficult to interpret or justify in real-world operational decision-making contexts [11]. Electricity planners and grid operators are therefore left with accurate predictions but no behavioral understanding of why demand behaves the way it does on specific days.

Clustering techniques have been applied in several studies to group users or load profiles, but in most cases clustering is used only as a preprocessing step to improve forecasting accuracy rather than as an independent mechanism for understanding daily demand behavior [4][7]. Studies that do apply clustering for load classification focus primarily on monthly peak-load grouping or industrial demand profiling and do not analyze day-level behavioral patterns within a national electricity system in a way that supports practical planning and operational decisions [3][6]. Similarly, anomaly detection frameworks in existing research are designed for fraud identification and abnormal consumption detection at the individual consumer or meter level and do not connect anomaly findings to broader system-level demand behavior analysis [2][5][9]. These approaches address individual aspects of electricity demand analysis in isolation without providing a comprehensive behavioral understanding of demand dynamics.

The most significant gap identified across all reviewed studies is the complete absence of a unified framework that integrates day-level demand pattern clustering, system-level anomaly

detection, peak demand day identification, and interpretable behavioral labeling within a single analytical system. No existing study or practical tool combines all of these capabilities in a way that transforms raw electricity demand data into actionable behavioral intelligence for electricity planning and grid management. This gap is directly confirmed by evidence from reviewed literature, where a comprehensive forecasting review identified interpretability and behavioral analysis as unaddressed weaknesses [11], a Sri Lanka specific forecasting study confirmed that existing local models do not incorporate behavioral profiling or demand characterization [13], and multiple clustering and anomaly detection studies demonstrated their contributions while remaining limited to narrow individual tasks without integration into a broader behavior intelligence approach [3][4][6][7][2][5][9].

To address this gap, this research proposes the development of a Behavior Intelligence Module as part of a Short-Term Load Forecasting system for Sri Lanka. The novelty of this research lies in shifting the focus of electricity demand analysis from purely numerical prediction toward behavior-based demand intelligence. The proposed module is the first framework designed for the Sri Lankan electricity context that combines demand pattern clustering using K-Means and DBSCAN, anomaly detection using Isolation Forest and Z-Score methods, and interpretable daily behavior labeling within a single unified pipeline. The creative contribution of this work is the behavioral taxonomy it introduces, classifying each demand day as a Normal Weekday Demand day, Peak Demand Day, Holiday Demand day, or Abnormal Demand Day, providing electricity planners and system operators with a clear and human-readable understanding of demand dynamics that no existing system currently offers.

2. OBJECTIVE

2.1 . Main Objective

The main objective of this research is to design, develop, and evaluate a Behavior Intelligence Module within a Short-Term Load Forecasting system for Sri Lanka that analyzes historical daily electricity demand patterns, detects anomalous demand events, and generates interpretable behavioral labels including Normal Weekday Demand, Peak Demand Day, Holiday Demand, and Abnormal Demand Day, using K-Means and DBSCAN clustering combined with Isolation Forest and Z-Score anomaly detection methods, in order to provide electricity system operators and energy planners with data-driven behavioral insights that support more informed operational and planning decisions, with the module fully developed, tested, and evaluated within the duration of this research project.

2.2. Specific Objectives

The following specific objectives have been defined to achieve the main research objective, with each objective aligned to an individual component of the proposed Behavior Intelligence Module:

2.2.1. To preprocess raw electricity demand data by handling missing values, removing abnormal measurements, and standardizing the dataset to ensure data quality for reliable behavioral analysis.

Raw electricity consumption data frequently contains missing readings, outliers, and inconsistent measurements that can negatively affect the accuracy of clustering and anomaly detection. Preprocessing ensures that the data fed into the system is clean, consistent, and representative of actual demand behavior, forming a solid foundation for all subsequent analytical components [11][13].

2.2.2. To aggregate cleaned time-series data into daily electricity demand profiles that capture each day's overall consumption pattern including peak periods, off-peak periods, and typical demand trends.

Since electricity consumption is recorded at 15-minute intervals, aggregating this data into daily profiles allows the system to capture the broader behavioral shape of each day. These daily profiles serve as the core input for clustering and behavior analysis, enabling the system to identify recurring demand patterns and day-level consumption dynamics [3][6].

2.2.3. To apply K-Means and DBSCAN clustering algorithms to group daily demand profiles with similar consumption characteristics and identify recurring demand patterns such as normal weekdays, weekends, holidays, and seasonal variations.

Clustering allows the system to discover natural groupings within historical demand data without requiring predefined labels. K-Means identifies groups with distinct demand levels while DBSCAN detects clusters of varying density and flags isolated unusual days, together

revealing the recurring behavioral structure of electricity demand across different day types [4][7].

2.2.4. To implement Isolation Forest and Z-Score anomaly detection methods to identify unusual demand events including peak demand spikes, abnormal consumption drops, and irregular days that deviate significantly from established behavioral patterns.

Detecting anomalies in electricity demand is critical for identifying unexpected events, system irregularities, and consumption inefficiencies. Isolation Forest isolates outlier demand profiles through random partitioning while Z-Score analysis flags days where aggregate consumption deviates significantly from the historical mean, enabling energy providers to respond proactively and improve grid reliability [2][5][9].

2.2.5. To generate interpretable behavior labels including Normal Weekday Demand, Peak Demand Day, Holiday or Low Demand Day, and Abnormal Demand Day for each identified demand cluster to provide operators with clear and actionable daily demand insights.

Converting clustered demand profiles into human-readable behavioral labels transforms raw data into operationally useful intelligence. These labels allow energy planners to quickly understand the behavioral character of any given day and make informed decisions regarding energy dispatch, load balancing, and contingency planning.

2.2.6. To evaluate the performance of the Behavior Intelligence Module by assessing clustering accuracy, anomaly detection precision, and the relevance of generated behavior labels against historical demand data and known real-world demand events.

Systematic evaluation ensures that the module produces outputs that are technically sound and practically meaningful for operational decision-making. By comparing module outputs against actual demand events and verified behavioral patterns in historical data, the research can objectively measure how effectively the system translates raw electricity demand data into actionable behavioral intelligence.

3.METHODOLOGY

This research adopts a Design Science Research (DSR) methodology to iteratively design, develop, and evaluate a Behavior Intelligence Module that analyzes daily electricity demand patterns within a Short-Term Load Forecasting system for Sri Lanka. Its goal is to provide electricity planners and system operators with transparent and actionable behavioral insights into daily electricity demand rather than simply delivering numerical forecast values. Quantitative measures will assess clustering accuracy, anomaly detection precision, and the relevance of generated behavior labels to validate the practical usefulness of the module.

3.1 Phase 1: Input Data and Preprocessing

The module relies on historical electricity demand data recorded at 15-minute intervals sourced from the Ceylon Electricity Board covering multiple years of consumption records. Input features include temporal indicators such as hour, day of week, and month, as well as calendar event flags identifying public holidays and seasonal periods. The preprocessing stage handles missing values through interpolation and forward-fill techniques, removes physically implausible readings and extreme outliers, and standardizes the dataset to ensure consistency and reliability. The cleaned data is then aggregated into daily electricity demand profiles, each represented as a sequence of 96 data points capturing quarter-hourly consumption throughout the day. These daily profiles serve as the fundamental input for all subsequent clustering and behavioral analysis components [11][13].

3.2 Phase 2: Clustering for Demand Pattern Discovery

Unsupervised machine learning clustering algorithms are applied to the daily demand profiles to identify groups of days sharing similar consumption characteristics. K-Means clustering is used to discover groups with distinct demand levels, with the optimal number of clusters determined using the Silhouette coefficient and Elbow method. DBSCAN is applied as a complementary density-based algorithm capable of detecting clusters of irregular shape and identifying isolated noise points representing unusual demand days. Together these methods reveal recurring behavioral patterns within the historical dataset including normal weekday demand, weekend demand, holiday demand, and seasonal demand variations. All clustering is performed using Python with Scikit-learn, Pandas, and NumPy libraries [4][7].

3.3 Phase 3: Anomaly Detection

The module implements two anomaly detection methods to identify demand days that deviate significantly from established behavioral patterns. Isolation Forest is applied to assign an anomaly score to each daily demand profile based on how easily it can be separated from the rest of the dataset through random partitioning. Z-Score analysis provides a complementary statistical measure that flags days where aggregate consumption deviates significantly from the historical mean. Days exceeding defined thresholds from either method are classified as

anomalous demand events. This stage enables electricity providers to detect unexpected consumption spikes, abnormal demand drops, and irregular operational conditions that would otherwise go unnoticed within routine forecasting outputs [2][5][9].

3.4 Phase 4: Behavior Label Generation

Each daily demand profile is assigned an interpretable behavior label derived from its cluster assignment, anomaly score, peak demand magnitude, and calendar context. The behavioral taxonomy classifies each day into one of the following categories: Normal Weekday Demand, Peak Demand Day, Holiday or Low Demand Day, and Abnormal Demand Day. These labels convert raw numerical demand data into human-readable behavioral intelligence that electricity planners and system operators can directly use without requiring deep technical knowledge of the underlying machine learning models. This label generation stage represents the core novelty of the proposed module, shifting electricity demand analysis from purely numerical prediction toward interpretable behavior-based intelligence.

3.5 Phase 5: Visualization and Output

The behavioral outputs of the module are presented through clear visualizations designed for electricity planners and grid operators. Visualizations include daily demand profile plots for each identified cluster, anomaly score distributions, behavior label summaries across the historical dataset, and calendar-based demand behavior maps showing how demand patterns vary across days, weeks, and seasons. These outputs allow system operators to quickly identify recurring demand patterns, flag unusual demand periods, and understand the behavioral character of any specific day within the historical record. All visualizations are implemented using Matplotlib and Seaborn within the Python development environment.

3.6 Phase 6: Validation Strategy

Validation ensures that the module produces outputs that are technically accurate, practically meaningful, and operationally useful. Clustering performance is evaluated using the Silhouette Score and Davies-Bouldin Index to measure how well daily demand profiles are grouped and how clearly separated the identified clusters are from one another. Anomaly detection performance is assessed by comparing detected anomalous days against verified historical records of known abnormal demand events such as extreme weather days, major public holidays, and documented load shedding periods, with Precision, Recall, and F1-Score used as evaluation metrics. The interpretability and usefulness of the generated behavior labels are verified through expert review conducted with electricity system operators and energy planners, who assess whether the assigned behavioral categories accurately reflect real-world demand characteristics and whether the outputs are sufficiently clear and actionable to support operational decision-making.

3.8. Tools and Technologies

Table 1 Tools and Technologies Used in the Behavior Intelligence Module

Category	**Technology**	**Justification**
Programming Language	Python	Widely adopted in electricity demand research, supports all required libraries, scalable for large time-series datasets, and ensures reproducibility
Data Handling	Pandas / NumPy	Efficient preprocessing, cleaning, aggregation, and manipulation of large 15-minute interval electricity demand datasets
Clustering Algorithms	Scikit-learn (K-Means / DBSCAN)	Proven unsupervised machine learning methods for grouping daily demand profiles and identifying recurring behavioral patterns
Anomaly Detection	Scikit-learn (Isolation Forest / Z-Score)	Effective and validated methods for detecting unusual demand events and flagging days that deviate significantly from established behavioral patterns
Behavior Labeling	Python / Pandas	Assigns interpretable behavioral categories to each demand day based on cluster assignment, anomaly score, peak magnitude, and calendar context
Cluster Validation	Scikit-learn (Silhouette Score / Davies-Bouldin Index)	Standard evaluation metrics for measuring clustering quality, internal cohesion, and cluster separation accuracy
Anomaly Evaluation	Scikit-learn (Precision / Recall / F1-Score)	Quantitative metrics for assessing anomaly detection accuracy against verified historical abnormal demand events
Visualization	Matplotlib / Seaborn	Generates daily demand profile plots, cluster distribution charts, anomaly score visualizations, and calendar-based behavior label maps for system operators
Development Environment	Jupyter Notebook / VS Code	Interactive development and testing environment suitable for iterative data analysis and module refinement

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a comprehensive electricity demand intelligence platform designed for Sri Lanka that integrates four core analytical services within a unified layered architecture. The platform combines behavioral demand analysis, short-term load forecasting, explainable artificial intelligence, and scenario-based decision support into a single connected system serving energy planners, Ceylon Electricity Board operators, and electricity analysts. The overall system architecture is illustrated in Fig. 1. The architecture is organized into five distinct layers: the client layer, the frontend layer, the backend layer, the AI layer, and the data layer, each with clearly defined responsibilities that communicate through secured and well-defined interfaces.

4.1 Component Overview

The Behavior Intelligence Service operates as the demand pattern analysis engine within the larger electricity demand intelligence platform. It interfaces directly with the Preprocessing and Feature Pipeline to receive cleaned and structured daily electricity demand profiles, and passes its behavioral outputs to the Forecast Service and the Application Database. The service produces three categories of output: clustered daily demand profiles that reveal recurring consumption patterns, anomaly scores that identify unusual demand events, and interpretable behavior labels that classify each day into a meaningful behavioral category. These outputs are consumed by energy planners and grid operators through the React.js frontend dashboard and are also utilized by the Forecast Service to enrich short-term load prediction with behavioral context, by the XAI Service to support explanation of detected demand patterns and anomalies, and by the Scenario Service to inform what-if demand scenario simulations and risk assessments. A Python-based backend served through the API Gateway handles all incoming requests to the Behavior Intelligence Service, providing real-time access to clustering outputs, anomaly detection results, behavior label summaries, and demand profile visualizations.

4.2 Module Structure

The Behavior Intelligence Service is structured across three main stages:

Input Stage: Receives cleaned and aggregated daily electricity demand profiles from the Preprocessing and Feature Pipeline, along with calendar context features including day of week, month, public holiday indicators, and seasonal flags sourced from the Time-Series Database.

Analysis Stage: Applies K-Means and DBSCAN clustering algorithms to group daily demand profiles with similar consumption characteristics, implements Isolation Forest and Z-Score anomaly detection to identify demand days that deviate significantly from established behavioral patterns, and generates interpretable behavior labels including Normal Weekday Demand, Peak Demand Day, Holiday or Low Demand Day, and Abnormal Demand Day for each day in the historical dataset.

Output Stage: Produces cluster assignment results, anomaly score distributions, behavior label summaries, and daily demand profile visualizations that are stored in the Application Database and served to the React.js frontend dashboard through the API Gateway for display to energy planners and system operators.

4.3 Data Flow

The data flow through the Behavior Intelligence Service follows a structured pipeline from raw data to interpretable behavioral intelligence:

1. Historical electricity demand data recorded at 15-minute intervals is retrieved from the Time-Series Database along with weather data, calendar information, and historical records.
2. The Preprocessing and Feature Pipeline cleans missing values, removes abnormal measurements, standardizes the dataset, and aggregates the cleaned time-series data into daily electricity demand profiles consisting of 96 data points per day.
3. The cleaned daily demand profiles and calendar context features are passed to the Behavior Intelligence Service through the shared backend processing layer.
4. K-Means and DBSCAN clustering algorithms are applied to group daily profiles with similar consumption characteristics and identify recurring demand patterns.
5. Isolation Forest and Z-Score anomaly detection methods are applied to each daily profile to generate anomaly scores and flag unusual demand events.
6. Interpretable behavior labels are assigned to each day based on cluster assignment, anomaly score, peak magnitude, and calendar context.
7. Clustering results, anomaly scores, and behavior labels are stored in the Application Database and passed to the Forecast Service as behavioral context inputs.
8. The React.js frontend retrieves behavior label summaries, cluster visualizations, and anomaly reports through the API Gateway and displays them to energy planners and system operators via interactive dashboards.

4.4 Integration with Other Components

The Behavior Intelligence Service exposes its analytical outputs to all other components within the AI layer through the shared backend processing layer. Behavior labels and cluster assignments are consumed by the Forecast Service to provide demand pattern context that improves short-term load prediction accuracy. Detected anomalies and behavior classifications are utilized by the XAI Service to support interpretable explanation of unusual demand patterns and peak demand events. The Scenario Service draws on behavior label outputs to support what-if demand scenario simulations and risk assessment activities for electricity planning. All components share a unified electricity demand dataset combined with calendar and contextual information stored in the centralized Time-Series Database, ensuring

consistency in data usage across behavior analysis, forecasting, explainability, and scenario simulation modules throughout the platform.

4.5 Scalability and Security Considerations

The Behavior Intelligence Service is designed with scalability and security as core architectural requirements. All API endpoints exposed by the service are secured through token-based authentication and HTTPS encryption managed by the API Gateway, ensuring that only authorized users and components can access behavioral intelligence outputs. Raw electricity demand data and historical records are processed entirely within the backend and are never exposed directly to the frontend. Only processed outputs including behavior labels, cluster summaries, anomaly scores, and visualizations are transmitted to the React.js dashboard, minimizing the risk of sensitive data exposure. The modular design of the service allows clustering algorithms, anomaly detection methods, and behavior labeling mechanisms to be updated or extended independently without affecting other components in the platform. The backend architecture supports horizontal scaling through containerization, allowing the service to process larger historical datasets and handle increased concurrent user requests as the platform expands over time.

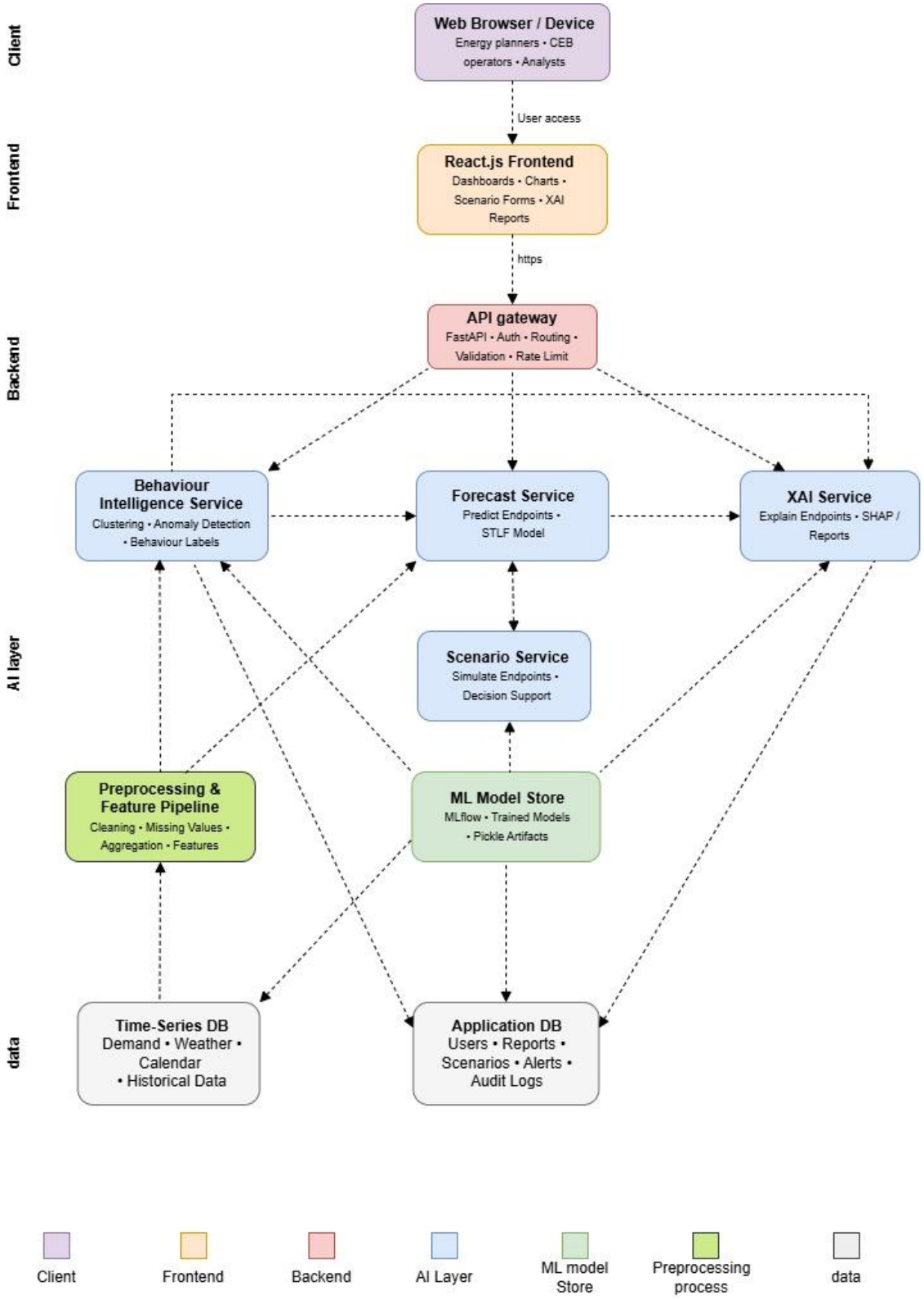


Figure 1High-Level System Architecture of the Proposed Electricity Demand Intelligence Platform

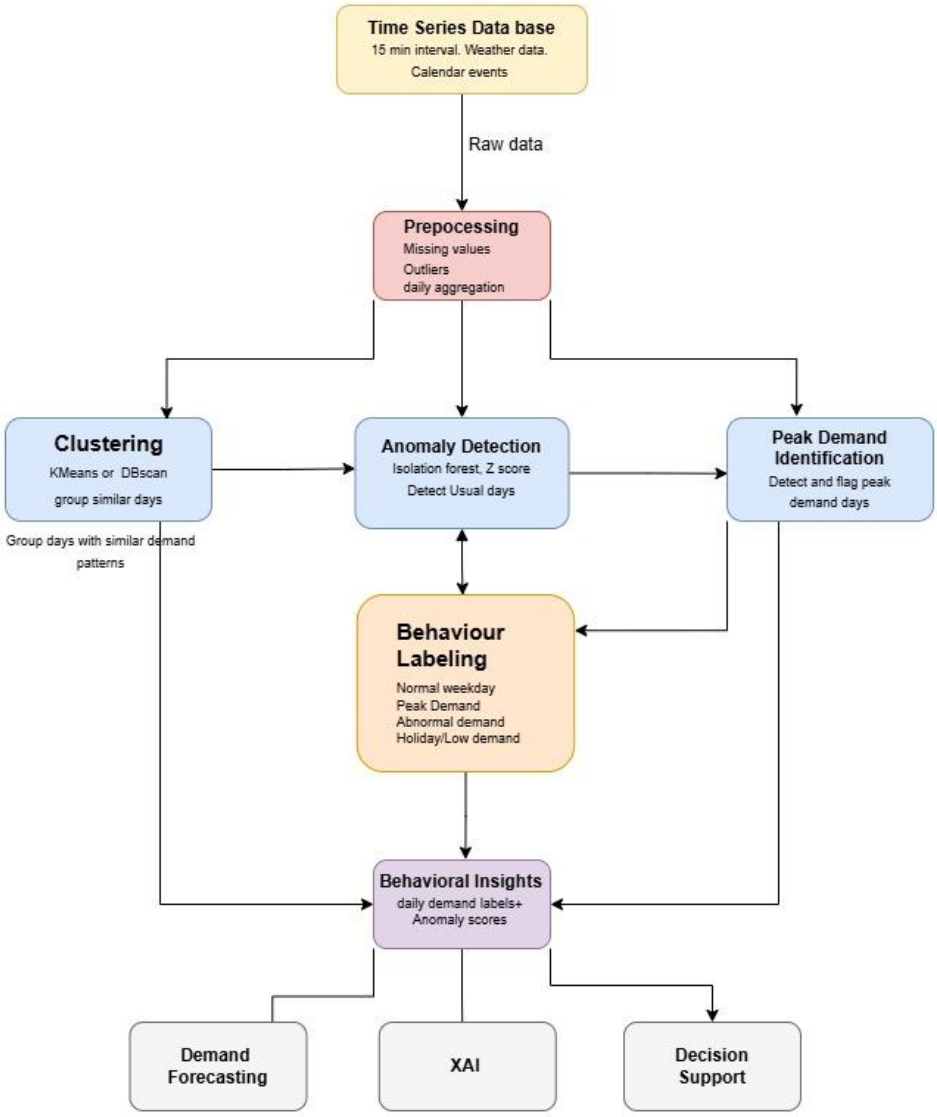


Figure 2 Behaviour Intelligence Module Individual Component Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Primary Users: Energy planners and system operators at the Ceylon Electricity Board who directly use the daily demand behavior insights, cluster summaries, anomaly reports, and behavior labels to plan electricity generation schedules and manage grid operations during peak demand periods.

Secondary Users: Grid managers and utility staff who access the behavior intelligence dashboards to monitor daily demand patterns, review flagged anomalous demand events, and make informed operational decisions regarding load balancing and energy dispatch.

Tertiary Stakeholders: Policymakers, regulatory authorities, and electricity demand researchers who review interpretable demand behavior insights to inform long-term energy policy decisions, infrastructure investment planning, and renewable energy integration strategies.

System Administrators: The SLIIT research team and future maintainers of the Behaviour Intelligence Module responsible for updating clustering models, refining anomaly detection thresholds, maintaining the data pipeline, and ensuring system reliability and scalability over time.

5.2 Functional Requirements

Table 2 Functional Requirements of the Behaviour Intelligence Module

FR#	Category	Requirement
FR1	Data Input	The system shall accept raw historical electricity demand data recorded at 15-minute intervals from the Ceylon Electricity Board along with weather data and calendar event information including public holidays and seasonal indicators.
FR2	Data Preprocessing	The system shall clean the raw input data by handling missing values through interpolation, removing physically implausible readings and extreme outliers, and standardizing the dataset to ensure data quality suitable for reliable behavioral analysis.
FR3	Daily Profile Generation	The system shall aggregate the cleaned time-series data into daily electricity demand profiles consisting of 96 data points per day, capturing quarter-hourly consumption patterns including peak periods, off-peak periods, and overnight demand trends.

FR4	Demand Day Clustering	The system shall apply K-Means and DBSCAN clustering algorithms to group daily demand profiles with similar consumption characteristics and identify recurring demand patterns such as normal weekdays, weekends, holidays, and seasonal variations.
FR5	Cluster Validation	The system shall evaluate clustering quality using Silhouette Score and Davies-Bouldin Index to determine the optimal number of clusters and ensure that identified demand pattern groups are internally coherent and well separated.
FR6	Anomaly Detection	The system shall implement Isolation Forest and Z-Score anomaly detection methods to identify daily demand profiles that deviate significantly from established behavioral patterns, including peak demand spikes, abnormal consumption drops, and irregular operational days.
FR7	Peak Demand Identification	The system shall detect and flag peak demand days based on anomaly scores, cluster characteristics, and peak magnitude thresholds, enabling electricity providers to respond proactively to high demand conditions.
FR8	Behavior Label Generation	The system shall assign interpretable behavior labels to each daily demand profile based on cluster assignment, anomaly score, peak magnitude, and calendar context. Labels shall include Normal Weekday Demand, Peak Demand Day, Holiday or Low Demand Day, and Abnormal Demand Day.
FR9	Behavioral Insight Output	The system shall generate actionable behavioral insights including daily demand label summaries, anomaly score distributions, cluster pattern reports, and calendar-based demand behavior maps that support electricity planning and grid management decisions.
FR10	Visualization Dashboard	The system shall present behavioral intelligence outputs through an interactive dashboard displaying daily demand profile plots, cluster distribution charts, anomaly score visualizations, behavior label summaries, and calendar-based demand behavior maps for energy planners and system operators.
FR11	Integration with Other Components	The system shall expose behavioral intelligence outputs including cluster assignments, anomaly scores, and behavior labels to the Forecast Service, XAI Service, and Scenario Service through the shared backend processing layer to support demand forecasting, explainability, and scenario simulation activities.
FR12	Backend API	The system shall provide REST API endpoints through the API Gateway for retrieval of clustering results, anomaly detection

		outputs, behavior label summaries, and demand profile visualizations by the React.js frontend dashboard.

5.3 Non-Functional Requirements

Table 3 Non-Functional Requirements of the Behaviour Intelligence Module

Category	Requirement
Performance	The system shall process historical electricity demand datasets containing over 189,888 records efficiently, with clustering and anomaly detection computations completing within acceptable response times to support near real-time operational decision-making.
Scalability	The system shall be scalable to accommodate growing data volumes from extended timeframes, increased data granularity, or additional data sources such as weather and economic indicators, without requiring significant architectural changes.
Usability	The behavior intelligence dashboard shall be intuitive and accessible to energy planners and grid operators with varying levels of technical expertise. Behavior labels, cluster summaries, and anomaly reports shall be presented in clear, human-readable formats that do not require knowledge of the underlying machine learning models.
Reliability	The system shall handle large datasets and varying input quality without crashes or failures. In the event of missing data or inconsistent inputs, the preprocessing pipeline shall apply appropriate imputation strategies and flag data quality issues without disrupting the overall behavioral analysis pipeline.
Interpretability	All behavioral outputs including cluster assignments, anomaly scores, and behavior labels shall be transparent and explainable, enabling system operators and energy planners to understand why a specific demand day has been assigned a particular behavioral category.
Security	All API communications between the backend and frontend shall be secured using TLS encryption and token-based JWT authentication. Raw historical electricity demand data shall not be transmitted to the frontend. Only processed outputs including behavior labels, aggregated statistics, and visualizations shall be shared with the dashboard, ensuring data confidentiality and compliance with organizational and national data privacy regulations.
Privacy	All electricity demand data used in this research shall be fully anonymized with no personally identifiable information included. Data handling shall comply with organizational data privacy policies and national regulations governing the use of utility consumption data.
Maintainability	The system shall be designed with a modular architecture that allows clustering algorithms, anomaly detection methods, and behavior labeling mechanisms to be updated, replaced, or extended independently without

	affecting other system components. Clear documentation shall be maintained for all modules to support future development and system enhancement.
Real-Time Operation	The system should support near real-time processing of incoming demand data to enable timely detection of anomalous demand events and rapid generation of behavior labels that can inform immediate operational decisions by grid operators.
Compatibility	The backend shall be compatible with Python 3.9 and above with all required libraries including Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn. The frontend dashboard shall be accessible on all modern web browsers without requiring additional software installation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary market for the Behaviour Intelligence Module targets electricity utilities and grid operators in Sri Lanka, with the Ceylon Electricity Board identified as the most direct potential adopter. The CEB manages a national grid serving approximately 7.2 million registered consumers and regularly experiences peak demand exceeding 2,400 MW, making intelligent demand behavior analysis an immediate operational necessity rather than a future consideration.

The secondary market includes the Public Utilities Commission of Sri Lanka and government ministries responsible for energy infrastructure planning, renewable energy policy, and long-term grid development. As Sri Lanka targets 70 percent renewable energy in its generation mix by 2030, understanding daily demand behavior patterns becomes increasingly critical for matching variable renewable generation with actual consumption.

The tertiary market covers large industrial and commercial electricity consumers, smart grid solution providers operating in the South Asian region, and international energy consultancies advising on developing-country grid modernization projects.

Primary Customer Persona: Pradeep is a 42-year-old Senior Load Dispatch Engineer at the CEB System Control Centre with 15 years of operational experience. Every day he reviews demand data and prepares generation schedules but receives only numerical forecasts with no explanation of why demand behaves differently on specific days. He needs a system that automatically classifies each day into a clear behavioral category, flags unusual demand events in advance, and provides interpretable outputs he can act on without needing deep technical knowledge of machine learning models.

6.2 Value Proposition

The Behavior Intelligence Module transforms raw electricity demand data into interpretable day-level behavioral intelligence that no existing forecasting system in Sri Lanka currently provides. For every day in the dataset the module produces a clear behavioral classification telling operators whether the day represents Normal Weekday Demand, a Peak Demand Day, a Holiday or Low Demand pattern, or an Abnormal Demand Day requiring immediate attention.

Three specific operational outcomes are delivered. First, proactive peak demand preparation by identifying Peak Demand Days in advance based on behavioral pattern matching, enabling more efficient generation scheduling and fuel procurement. Second, early anomaly warning through Isolation Forest and Z-Score detection that flags unusual demand behavior before it escalates into grid instability. Third, interpretable operational intelligence where every

behavioral label is traceable to specific cluster characteristics and calendar context that operators can verify and act on with confidence.

6.3 Revenue Model

Software Licensing: Annual licensing for electricity utilities and energy planning agencies at LKR 250,000 to LKR 450,000 per institution per year based on data volume and number of users.

Consultancy and Customization: Project-based services for configuring the module to specific operational environments, ranging from LKR 500,000 to LKR 2 million per engagement.

SaaS Deployment: Cloud-hosted subscription plans for smaller utilities and research institutions ranging from LKR 15,000 to LKR 80,000 per month based on feature access level.

Research Partnerships: Licensing of clustering frameworks and behavior labeling methodologies to academic institutions and South Asian smart grid solution providers for collaborative research and development.

6.4 Cost Estimation

Initial development costs are estimated at LKR 17 million covering pipeline development at LKR 6.5 million, dashboard and labeling framework at LKR 4.2 million, data integration and preprocessing infrastructure at LKR 2.8 million, and cloud deployment setup at LKR 3.5 million.

Ongoing annual costs are estimated at LKR 12.3 million covering cloud hosting at LKR 2.4 million, technical staffing at LKR 7.2 million, quality assurance at LKR 1.5 million, and marketing at LKR 1.2 million per year.

Projected Year 1 revenue of LKR 7.8 million is expected from two utility licensing agreements, one CEB consulting engagement, one government research grant, and one regional licensing agreement. Break-even is projected by Year 3 as the licensed client base scales to five or more institutions.

6.5 Pricing Strategy

Regional Utility Plan at LKR 2 million per year covers core clustering, anomaly detection, behavior labeling, and standard dashboard access for smaller regional utilities.

Standard Utility Plan at LKR 4.5 million per year provides full clustering with seasonal analysis, complete anomaly detection, peak demand identification, interactive dashboards, and quarterly model recalibration for medium-sized utilities.

Enterprise Plan uses customized pricing for large-scale utilities and government agencies requiring full system integration, real-time processing, dedicated support, and custom behavior label taxonomies aligned with their operational workflows.

6.6 Competitive Advantage

No existing commercial product in Sri Lanka provides an integrated system combining day-level demand clustering, system-level anomaly detection, peak demand identification, and interpretable behavioral labeling within a single unified framework. The module is purpose-built for the Sri Lankan electricity context using CEB consumption data and local calendar patterns, making it immediately relevant in a way that generic international forecasting tools are not. Its modular integration architecture connects directly with the broader STLF platform without requiring clients to replace existing systems, and its alignment with Sri Lanka's 2030 renewable energy targets and United Nations Sustainable Development Goal 7 positions it as a strong candidate for government-backed energy modernization funding.

6.7 Intellectual Property Considerations

The primary intellectual property assets include the behavioral demand taxonomy designed for the Sri Lankan electricity context, the combined clustering and anomaly detection pipeline optimized for 15-minute interval national grid data, and the behavior label assignment methodology integrating cluster characteristics, anomaly scores, and Sri Lanka-specific calendar context. All software code, dashboard design, and backend API architecture developed during this research will be protected under copyright. Novel algorithmic configurations will be evaluated for patent registration, and commercial licensing agreements will be explored with Sri Lankan utilities, South Asian smart grid providers, and international energy management consultancies.

7. BUDGET AND JUSTIFICATION

The following budget outlines the estimated costs required to design, develop, and evaluate the Behaviour Intelligence Module within the Short-Term Load Forecasting system for Sri Lanka. All costs are estimated based on current Sri Lankan market rates for hardware, software, cloud services, and human effort as of 2025.

Table 4 Budget and Justification for the Behaviour Intelligence Module

Item	Qty	Unit (LKR)	Total (LKR)	Justification
High Performance Laptop for Clustering and Anomaly Detection	1	185,000	185,000	Required for executing K-Means and DBSCAN clustering algorithms on large daily demand profile datasets and running Isolation Forest and Z-Score anomaly detection computations locally during development and testing phases
External SSD Storage Drive	1	18,500	18,500	Dedicated local storage for raw CEB electricity demand datasets, preprocessed daily profiles, clustering model outputs, anomaly score records, and behavior label results to ensure fast data access during development
Cloud GPU Instance (AWS / Google Cloud)	8 months	6,800	54,400	GPU-accelerated cloud computing for processing over 189,888 historical demand records, executing batch clustering runs, and validating anomaly detection performance across multiple algorithm configurations
Backend Hosting Server	10 months	3,200	32,000	Cloud server deployment for hosting the Behaviour Intelligence API endpoints that serve clustering results, anomaly scores, and behavior label outputs to the frontend dashboard through the API Gateway
Domain Registration and SSL Certificate	1	7,500	7,500	Secure domain and HTTPS certificate for the Behaviour Intelligence dashboard ensuring all behavioral intelligence outputs transmitted between backend and

				frontend are protected from unauthorized interception
Cloud Data Backup and Storage Service	10 months	2,200	22,000	Reliable cloud backup for historical CEB demand datasets, preprocessed daily demand profiles, trained clustering models, anomaly detection configurations, and generated behavior label archives throughout the research period
Research Data Collection and CEB Coordination	4 visits	4,500	18,000	Transportation and coordination costs for four visits to CEB facilities and relevant government energy institutions for dataset access negotiations, stakeholder consultations, and data quality verification meetings
Visualization and Dashboard Development Tools	1	8,500	8,500	Software tools and library configuration costs for building the interactive behavior intelligence dashboard including daily demand profile plots, cluster distribution charts, anomaly score visualizations, and calendar-based behavior label maps
Research Documentation and Report Preparation	1	12,500	12,500	Costs covering academic report printing, research proposal documentation, conference paper preparation, internet access for literature review, and stationery expenses throughout the full project duration
Contingency Reserve	1	15,000	15,000	Reserved budget allocation for unexpected expenses including additional cloud computing requirements during peak testing phases, software compatibility issues, or additional data coordination visits beyond the planned schedule
TOTAL BUDGET			**LKR 373,400**	

Budget Summary by Category:

Category	Total (LKR)
Hardware Costs	203,500
Cloud Services	108,400
Software and Development Tools	8,500
Human Effort and Data Collection	18,000
Documentation and Miscellaneous	27,500
Contingency Reserve	15,000
Grand Total	**LKR 373,400**

The hardware investment of LKR 203,500 covering the laptop and external SSD drive forms the core local development environment necessary for iterative clustering experiments and anomaly detection testing without depending entirely on cloud resources. Cloud GPU computing costs of LKR 54,400 over eight months provide the processing power required to handle the full scale of CEB historical demand data efficiently, particularly during batch clustering validation runs and anomaly detection threshold calibration. Backend hosting costs of LKR 32,000 over ten months ensure reliable deployment of the API infrastructure connecting the analytical pipeline to the frontend dashboard throughout the evaluation period. The domain and SSL investment of LKR 7,500 guarantees secure communication of behavioral intelligence outputs between system components in compliance with data security requirements. Cloud backup and storage costs of LKR 22,000 protect all research data assets and model outputs against loss throughout the project lifecycle. Research data collection costs of LKR 18,000 cover four planned coordination visits to CEB facilities for dataset procurement and stakeholder engagement activities. Development tool costs of LKR 8,500 cover the complete Python analytical environment configuration including Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn setup. Documentation costs of LKR 12,500 cover all report preparation and academic communication requirements. The contingency reserve of LKR 15,000 provides financial flexibility to address unexpected technical or logistical challenges without disrupting the research timeline.

8. WORK BREAKDOWN STRUCTURE (WBS)

Table V presents the complete work breakdown structure and project timeline for the Behaviour Intelligence Module, outlining all tasks, estimated durations, deliverables, and current completion status across the individual research project timeline.

Table 5 Work Breakdown Structure and Timeline for the Behaviour Intelligence Module

Task ID	Task Name	Key Activities	Est. Duration	Week	Deliverable	Status
T1	Research and Requirements Analysis	Literature review, stakeholder identification, requirement gathering, dataset assessment, behavior label taxonomy definition	2 weeks	Week 1 to 2	Requirements specification document and finalized behavior label taxonomy	Completed
T2	Data Collection and Preprocessing	CEB dataset procurement, missing value handling, outlier removal, data standardization, calendar feature encoding	2 weeks	Week 3 to 4	Cleaned and standardized electricity demand dataset ready for analysis	Completed
T3	Daily Demand Profile Generation	15-minute data aggregation into 96-point daily profiles, feature extraction including peak magnitudes and variability metrics, profile validation	1 week	Week 5	Validated daily demand profile dataset with extracted features	In Progress
T4	Clustering Implementation	K-Means implementation, DBSCAN implementation, optimal	3 weeks	Week 6 to 8	Clustered daily demand profiles with optimal grouping	Planned

		cluster determination using Silhouette and Elbow method, algorithm comparison			results and evaluation metrics	
T5	Anomaly Detection Implementation	Isolation Forest implementation, Z-Score analysis, anomaly threshold calibration, validation against historical CEB records	2 weeks	Week 9 to 10	Anomaly score dataset with flagged irregular demand days verified against real events	Planned
T6	Behavior Label Generation	Label assignment mechanism development, behavioral category assignment for all demand days, accuracy verification against CEB records, label refinement	2 weeks	Week 11 to 12	Complete behavior labeled dataset with Normal Weekday, Peak Demand, Holiday, and Abnormal Day classifications	Planned
T7	Visualization and Dashboard Development	Interactive dashboard design, demand profile plots, cluster charts, anomaly visualizations, calendar behavior maps, user control implementation	2 weeks	Week 13 to 14	Fully functional interactive Behaviour Intelligence dashboard	Planned
T8	Backend API Development	FastAPI backend setup, REST endpoint implementation, integration with Forecast Service and XAI Service,	2 weeks	Week 15 to 16	Deployed and secured backend API serving all behavioral intelligence outputs	Planned

		TLS and JWT security configuration				
T9	Validation and Performance Evaluation	Silhouette Score and Davies-Bouldin Index evaluation, Precision Recall and F1-Score assessment, expert review with CEB operators, feedback incorporation	2 weeks	Week 17 to 18	Validated module with full performance evaluation report and expert review findings	Planned
T10	Documentation and Submission	Technical report writing, API documentation, research paper draft preparation, source code documentation, final project submission	2 weeks	Week 19 to 20	Final research report, API documentation , research paper draft, and complete project submission package	Planned

Total Project Duration: 20 Weeks

Phase	Tasks Covered	Duration
Phase 1 — Foundation	T1, T2, T3	Week 1 to 5
Phase 2 — Core Development	T4, T5, T6	Week 6 to 12
Phase 3 — System Build	T7, T8	Week 13 to 16
Phase 4 — Evaluation and Submission	T9, T10	Week 17 to 20

9. GANTT CHART

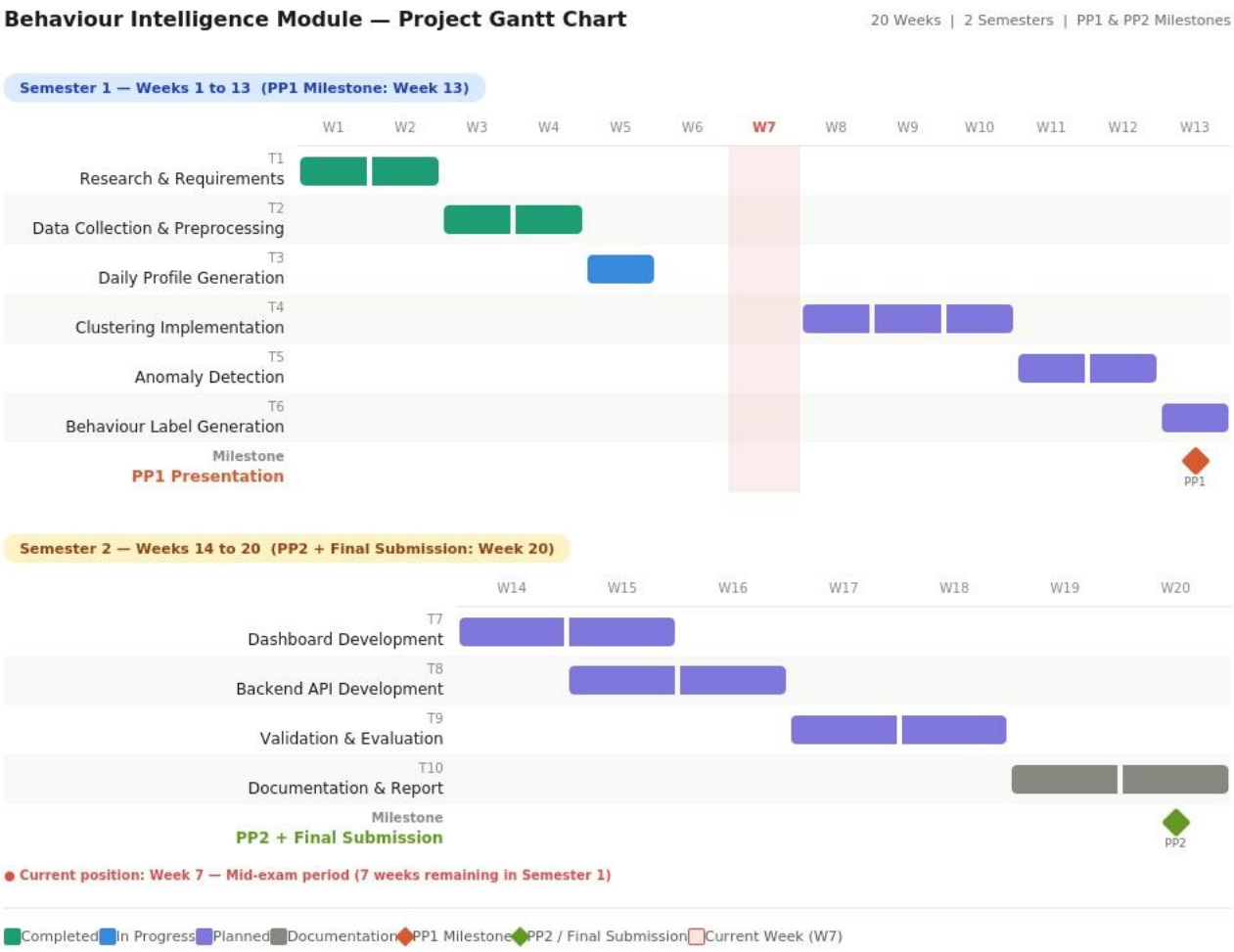


Figure 3 Project Gantt Chart — Semester-wise Timeline with Milestones

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11. APPENDICES

Appendix A: Risk Analysis

The following table identifies the key risks associated with the development and evaluation of the Behaviour Intelligence Module, along with their likelihood, potential impact, and proposed mitigation strategies.

Risk	Likelihood	Impact	Mitigation Strategy
Insufficient historical electricity demand data from CEB covering multiple years at 15-minute intervals	Medium	High	Explore supplementary publicly available Sri Lankan electricity datasets and apply data augmentation techniques to fill coverage gaps
Poor clustering quality resulting in overlapping or meaningless demand day groupings	Medium	High	Apply multiple cluster validation metrics including Silhouette Score and Davies-Bouldin Index and compare K-Means and DBSCAN outputs before finalizing clusters
Anomaly detection producing excessive false positives or missing genuine anomalous demand events	Medium	High	Calibrate Isolation Forest and Z-Score thresholds against verified historical abnormal demand records from CEB operational reports
Behaviour labels being incorrectly assigned due to ambiguous cluster boundaries or overlapping demand profiles	Medium	High	Conduct expert review with CEB system operators to validate label accuracy and refine assignment logic based on operational feedback
Missing or inconsistent calendar and public holiday data affecting clustering and labeling accuracy	Medium	Medium	Cross-reference multiple official Sri Lanka government calendar sources and manually verify public holiday records for the full dataset period
Computational performance issues when processing large datasets containing over 189,888 demand records	Low	Medium	Optimize preprocessing pipeline and use batch processing with cloud GPU resources through Google Colab Pro to handle large-scale computations efficiently
Integration issues between the Behaviour Intelligence Service and other platform components including Forecast Service and XAI Service	Medium	Medium	Follow agreed API specifications and conduct joint integration testing with group members responsible for other components during the backend development phase
Misinterpretation of behaviour labels by energy planners	Medium	Medium	Design dashboard outputs in plain human-readable language

unfamiliar with machine learning outputs			and conduct usability testing with target users to ensure clarity and accessibility
Climate or seasonal changes affecting demand pattern consistency between training and testing periods	Low	Medium	Use multi-year historical data spanning diverse seasonal periods and validate clustering outputs separately for each season

Appendix B: Ethical Considerations

The following ethical principles and safeguards have been adopted throughout the design and development of the Behaviour Intelligence Module to ensure responsible and compliant research practices.

All historical electricity demand data used in this research is sourced from the Ceylon Electricity Board and publicly available datasets in aggregated form. No personally identifiable information is collected, processed, or stored at any stage of the research. All data handling follows responsible usage practices that ensure transparency, integrity, and accountability in all analytical processes.

Behavioural outputs including cluster assignments, anomaly scores, and demand labels are generated at the system level and do not identify, profile, or expose individual electricity consumers. All visualizations and dashboard outputs are presented in aggregate form consistent with consumer privacy protection requirements.

Ethical approval will be sought from the SLIIT Ethics Review Committee prior to commencing data collection and analysis activities to ensure full compliance with institutional research ethics guidelines. All research activities are conducted in alignment with Sri Lanka's Personal Data Protection Act No. 9 of 2022, ensuring lawful and responsible use of electricity consumption data throughout the project.

Clustering algorithms, anomaly detection methods, and behaviour labeling mechanisms are designed and validated to avoid producing misleading or harmful outputs. Clear explanations accompany all system outputs to prevent misinterpretation by electricity planners and system operators. Cultural, social, and operational contexts specific to Sri Lanka's electricity system are carefully considered to ensure that all analytical outputs and behavioral classifications are relevant, appropriate, and practically useful for the intended stakeholders.

Appendix C: Technology and Framework Stack

The following tools and technologies are used in the implementation of the Behaviour Intelligence Module:

Python 3.11+: Core programming language for all data preprocessing, feature engineering, clustering implementation, anomaly detection, and behaviour label generation activities throughout the module.

Pandas and NumPy: Data handling and numerical computation libraries used for loading, cleaning, transforming, and aggregating large 15-minute interval electricity demand time-series datasets into daily demand profiles.

Scikit-learn: Machine learning library providing implementations of K-Means clustering, DBSCAN, Isolation Forest, Silhouette Score, Davies-Bouldin Index, and performance evaluation metrics including Precision, Recall, and F1-Score.

Matplotlib and Seaborn: Visualization libraries used for generating daily demand profile plots, cluster distribution charts, anomaly score visualizations, and calendar-based behaviour label maps for the interactive Behaviour Intelligence dashboard.

FastAPI: Backend API framework for exposing REST endpoints that serve clustering results, anomaly detection outputs, and behaviour label summaries to the React.js frontend dashboard through the shared API Gateway.

React and JavaScript: Frontend framework for building the interactive Behaviour Intelligence dashboard through which energy planners and system operators access daily demand pattern summaries, anomaly reports, and behaviour label outputs.

SQLite and PostgreSQL: Database systems for storing preprocessed daily demand profiles, clustering model outputs, anomaly scores, behaviour label records, and system audit logs throughout the research and deployment phases.

Google Colab Pro: Cloud-based GPU computing environment used for processing large historical demand datasets, running batch clustering experiments, and validating anomaly detection performance across multiple algorithm configurations.

Git and GitHub: Version control and collaborative development platform ensuring reproducible and well-documented module development, testing, and integration activities.

Jupyter Notebook and VS Code: Development environments used for iterative data analysis, algorithm experimentation, clustering visualization, and anomaly detection threshold calibration during the research phase.

Appendix D: Preliminary UI Mockups — Behaviour Intelligence Dashboard

The Behaviour Intelligence dashboard is designed to present daily electricity demand behavioral outputs in a clear, accessible, and operationally useful format for energy planners and system operators at the Ceylon Electricity Board. The dashboard interface incorporates the following key elements:

A daily demand behavior summary panel displayed at the top of the screen showing the assigned behavior label for the selected date, for example Normal Weekday Demand, Peak Demand Day, Holiday or Low Demand Day, or Abnormal Demand Day, along with the corresponding anomaly score and cluster assignment identifier.

A daily demand profile visualization displaying the 96-point quarter-hourly consumption curve for the selected day, overlaid with the average profile of its assigned cluster to highlight how the selected day compares to its behavioral group.

A cluster distribution chart showing how all historical demand days are distributed across identified behavioral clusters, allowing operators to understand the frequency and seasonal patterns of each demand category.

An anomaly score panel displaying the Isolation Forest and Z-Score anomaly scores for the selected day, with a clear visual indicator showing whether the day falls within normal behavioral bounds or has been flagged as an anomalous demand event.

A calendar-based behaviour label map providing a month-view visualization of behavior labels across all historical dates, enabling operators to quickly identify recurring peak demand periods, seasonal patterns, and clusters of abnormal demand days.

A date and filter selector allowing users to navigate to any specific date in the historical dataset, filter by behavior label category, or view demand patterns within a selected date range for operational planning purposes.

A textual behavioral insight panel generating a plain language summary such as: "This day was classified as a Peak Demand Day. Consumption exceeded the normal weekday cluster average by 18 percent and the anomaly score indicates significantly elevated demand relative to historical patterns for this calendar period."

An integration status panel showing the behavioral intelligence outputs that have been passed to the Forecast Service, XAI Service, and Scenario Service, confirming that behavioral context from the current analysis has been successfully shared across all connected platform components.

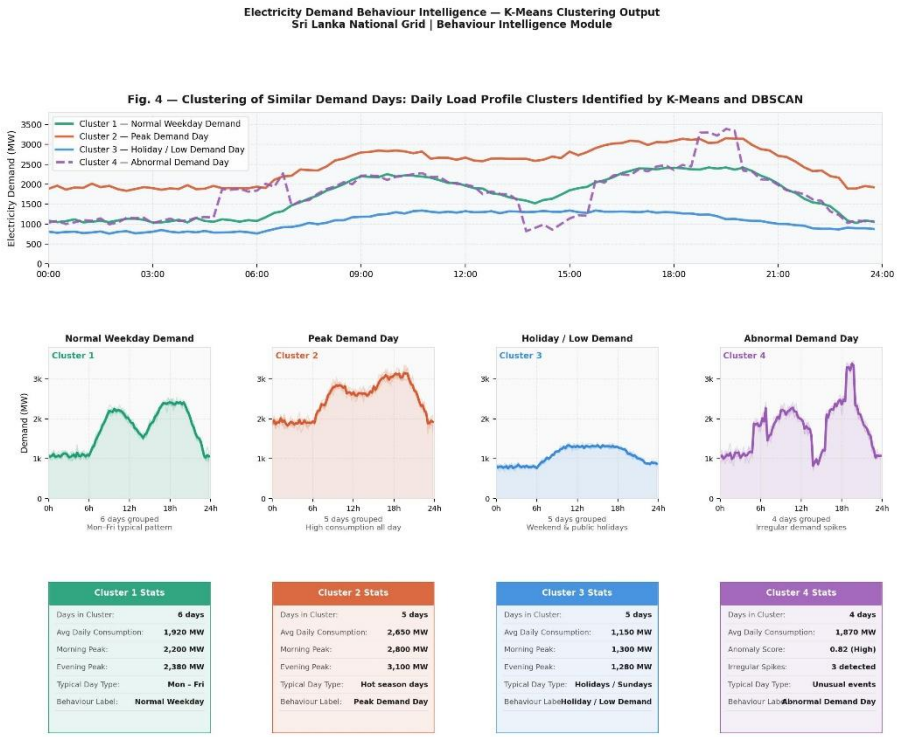


Figure 4 Clustering of Similar Demand Days: Daily Load Profile Clusters Identified by K-Means and DBSCAN